

Supplemental Materials for “Automated severe aortic stenosis detection on single-view echocardiography: A multi-center deep learning study”

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Supplemental methods

Echocardiogram interpretation & labelling: The presence of AS severity was adjudicated based on the original echocardiographic report and reflected the final severity grade assigned by the interpreting physician. As a general rule, Doppler-based indices were interpreted based on the parameters recommended by the American Society of Echocardiography (ASE), which included peak aortic valve (stenosis) jet velocity, mean transaortic/trans-valvular gradient, and mean valve area, as assessed by the continuity equation. According to the guidelines, cut-offs of >4 m/sec, >40 mm Hg and less than <1.0 cm², respectively, were consistent with severe AS.¹ The left ventricular ejection fraction (LVEF) was reported based on three-dimensional (3D) echocardiography, and in the absence of that, based on the Simpson's biplane method. In the absence of these measurements, we reported the lower end of the visually estimated LVEF. Since severe AS detection was formulated as a binary classification task, all AS designations other than "severe AS" were binned into the "not severe AS" category.

Data extraction and de-identification: In the New England cohort, after excluding studies that were not properly extracted from the database, 13,354 studies first underwent de-identification. After loading the pixel data for each video with the pydicom library (<https://pydicom.github.io/>), pixels in the periphery of each video frame were masked out to remove identifying information, and videos were converted to the Audio Video Interleave (AVI) format to enable fast loading for later preprocessing steps.

View classification: The resulting 539,188 videos from 12,185 studies then underwent view classification. Using the pretrained TTE view classifier from Zhang *et al.*,² ten frames from each de-identified video were randomly selected, downsampled to 224 x 224 resolution, and fed through the pretrained VGG19 convolutional neural network. Video-level view predictions were then obtained by averaging each video's 10 frame-wise view probabilities, and videos that were most confidently predicted as PLAX were kept for further preprocessing. While the pretrained view classifier was capable of discriminating variants of the standard PLAX view such as "PLAX", "PLAX – remote," "PLAX – zoom of left atrium," and "PLAX – centered over left atrium," we elected to only proceed with videos most confidently classified as "PLAX." To manually verify the accuracy of view classification, we assessed a subset of 50 videos that were predicted to be PLAX and found that only 1 out of 50 (2%) was incorrect. These 50 videos were randomly sampled from each predicted probability decile (i.e., 5 videos with a predicted PLAX probability of 0-10%, 5 videos with a predicted PLAX probability of 10-20%, and so on). This suggests that even when a video is not confidently classified as PLAX, the classifier is still almost always correct.

Initial video pre-processing: After view classification, the 37,788 videos from 11,480 studies were prepared for deep learning model development. Given differences in AS severity measures across different domains, we excluded echocardiograms with low-flow, low-gradient & paradoxical aortic stenosis leaving 37,611 PLAX videos from 11,424 studies. All videos underwent a more thorough cleaning and de-identification process that involved binarizing each video frame with a fixed threshold, then masking out all pixels outside the convex hull of the largest contour in order to remove all information outside the central image content. Finally, each

video clip was spatially downsampled to 112 x 112 and saved to AVI format for fast loading during model training.

Self-supervised contrastive learning: To learn transferable representations of PLAX echocardiogram videos for downstream severe AS identification, we performed self-supervised pretraining on all training set videos. This pretraining step critically enables the model to learn representations of echocardiograms that are robust to standard variations in video acquisition, thus better generalizing when later fine-tuned on a specific downstream task. We have previously demonstrated that a more appropriate initialization for data-efficient classification tasks could be achieved by “in-domain” pretraining on echocardiograms,³ as opposed to other standard approaches such as random initialization of weights and transfer learning.⁴⁻⁶ To this end, we have designed a novel self-supervised learning algorithm specifically catered to echocardiogram videos. More specifically, self-supervised representation learning was performed on the training set videos with a novel combination of (i) a multi-instance contrastive learning task and (ii) a frame re-ordering pretext task. In brief, we adopted the SimCLR framework⁷ for contrastive learning, which traditionally generates two “views” of an image by sending two copies of the input through a pipeline of random image augmentations, producing view $\tilde{x}_i$ and $\tilde{x}_j$. An encoder $f()$ is then used to learn representations of each view, $h_i = f(\tilde{x}_i)$ and $h_j = f(\tilde{x}_j)$, which are then projected to a lower dimensionality with a projector $g()$. The resulting learned embeddings of each view, $z_i = g(f(\tilde{x}_i))$ and $z_j = g(f(\tilde{x}_j))$ are then “contrasted” via the temperature-normalized cross-entropy (NT-Xent) loss, which encourages the model to learn *similar* representations of views from the same original image (so-called “positive pairs”) and *dissimilar* representations of views from all other images (“negative pairs”) in a given minibatch. The model is then trained to learn *similar* representations of views from the same original image (so-called “positive pairs”) and *dissimilar* representations of views from all other images.

While SimCLR has proven very successful for 2D natural images as well as in medical applications such as radiography and dermatological images, there are several barriers to its successful adaptation to echocardiogram videos. First, SimCLR requires extremely heavy image augmentation for effective representation learning, which would destroy valuable signal encoded in the brittle, noisy ultrasound images produced by echocardiography. Second, SimCLR was designed for 2D images, which would completely ignore the temporal dimension of echocardiography. To address the first issue, we utilized “multi-instance” contrastive learning – borrowing language and key insights from Azizi *et al.*⁸ – whereby we form positive pairs between *different* videos from the *same* patient. This critically removes the need to synthetically create two different “views” of a patient by heavily augmenting their echo video, instead leveraging the fact that almost all studies contain multiple distinct PLAX videos of a patient.

To address the second issue, we additionally included a frame re-ordering “pretext” task to our self-supervised learning method, where we randomly permuted the frames of each input echo, then trained the model to predict the original order of frames. Similar to the approach of Jiao *et al.*,⁹ this frame re-ordering task is treated as a classification problem and was implemented with a simple fully-connected layer that minimizes the cross-entropy between the known and predicted original frame order; specifically, if an input echo clip has K frames, then the $K!$ possible permutations of frames served as the targets for classification. Then the loss function of our self-supervised learning method is simply the sum of the contrastive NT-Xent objective and the pretext frame re-ordering cross-entropy objective. Self-supervised pretraining was performed on randomly sampled video clips of 4 consecutive frames from each of the

training set echocardiogram videos for 300 epochs. The encoder $f()$ was a randomly initialized 3D-ResNet18,¹⁰ and the projector $g()$ projected each 512-dimensional learned representation down to a 128-dimensional representation with a hidden layer of 256 units followed by a ReLU activation, followed by an output layer with 128 units. The model was trained for 300 epochs on all unique pairs of *different* PLAX videos from the same patient with the Adam optimizer¹¹ and a learning rate of 0.1, a batch size of 392 (196 per GPU), and NT-Xent temperature hyperparameter of 0.5. The following augmentations were applied to each frame in a temporally consistent manner (same transformations for each frame of a given video clip): random zero padding by up to 8 pixels in each spatial dimension, a random horizontal flip with probability 0.5, and a random rotation within -10 and 10 degrees with probability 0.5. After augmentation, each video clip was normalized so that the maximum pixel intensity was mapped to 1 and the minimum intensity to 0. The SSL model was trained on two NVIDIA RTX 3090 GPUs.

Deep neural network training for severe AS prediction: The same 3D-ResNet18 architecture was used to predict severe AS. Three different methods were used to initialize the parameters of this network: an SSL initialization, a Kinetics-400 initialization, and a random initialization. The SSL initialization directly used the learned weights of the encoder from the SSL pretraining step described in detail above. The Kinetics-400 initialization represents the “standard” transfer learning approach for 3D data, using the weights from a 3D-ResNet18 trained in a supervised fashion on the Kinetics-400 dataset, a large corpus of over 300,000 natural videos for human action classification; these weights are readily available through the torchvision API (<https://pytorch.org/vision/stable/index.html>) provided by PyTorch. The random initialization is the default when initializing a 3D-ResNet18 with PyTorch.¹²

All fine-tuning models were trained on randomly sampled video clips of 16 consecutive frames from training set echocardiograms, optionally padding with empty frames along the temporal axis if either the video is too short or the randomly chosen starting point of the clip is near the end of the video. The same augmentations were used as in self-supervised pretraining, and all video clips were min-max normalized; when fine-tuning from a Kinetics-400 initialization, video clips were further standardized using the channel-wise means and standard deviations from the Kinetics-400 training dataset, a standard preprocessing step when performing transfer learning. All models were trained for a maximum of 30 epochs with early stopping – specifically, if the validation AUROC did not improve for 5 consecutive epochs, training was terminated and the weights from the epoch with maximum validation AUROC were used for final evaluation. Severe AS models were trained on a single NVIDIA RTX 3090 GPU with the Adam optimizer, a learning rate of 1×10^{-4} (except the SSL-pretrained model, which used a learning rate of 0.1), and a batch size of 88 in order to maximize GPU utilization. Since this problem was framed as a binary classification task, these models minimized a sigmoid cross-entropy loss. We additionally used class weights computed with the method provided by scikit-learn¹³ to accommodate class imbalance in addition to label smoothing^{14,15} with $\alpha=0.1$, a method to improve model calibration and generalization. Learning curves depicting loss throughout training were graphically visualized.

Ensembling: We formed an ensemble of three models trained to detect severe AS, with diversity injected by the dramatically different methods used to initialize each model’s weights before training. Ensembling is known to improve predictive performance by aggregating the outputs of

multiple independently trained models.¹⁶ Moreover, statistical^{17–19} and deep learning^{20–22} studies have shown that ensembles of *diverse* constituent models are most effective.

More specifically, since the fine-tuned severe AS models are trained on 16-frame video clips, yet AS labels describe each study, we first aggregated clip-level predictions into study-level predictions for performance evaluation. When performing inference on an echo video, four evenly spaced clips (potentially with overlapping frames) of 16 consecutive frames were extracted and fed into the trained AS model. These clip-level predictions were then averaged to obtain a video-level prediction of severe AS. After repeating this process for all videos, the severe AS probabilities for videos in each study were averaged to obtain study-level AS predictions that could be used to compute evaluation metrics. The final ensemble model is then formed by averaging the output probabilities of the SSL-pretrained model, the Kinetics-400-pretrained model, and the randomly initialized model after fine-tuning each ensemble member to classify severe AS. Since no quality control is applied when selecting PLAX videos for this work, averaging results over multiple videos in the same study has a stabilizing effect that boosts predictive performance.

Model explainability: Saliency maps were generated by leveraging the Grad-CAM method²³ for obtaining visual explanations from deep neural networks. In simple terms, saliency maps refer to a visualization technique that highlights the specific parts of an image that contribute the most to the output of a given model. This is achieved by computing the gradient of the model output relative to each input image, and then assigning importance scores to each pixel of the original image. Pixels with higher scores are therefore considered important to the model's final prediction. Specifically, in our study, heatmaps were generated by applying Grad-CAM to a clip of the first 32 frames of an echo, using the last convolution block of the 3D ResNet18 to generate a 7 x 7 x 4 (height x width x time) heatmap displaying roughly where the model is attending to over the spatial and temporal dimensions. The Grad-CAM output was interpolated to the original input dimension of 112 x 112 x 32 with the scipy “zoom” function; this process produces a frame-by-frame “visual explanation” of *where* the model is focusing frame by frame in order to make its prediction. However, to generate a single 2D heatmap for a given echo clip, the pixelwise maximum along the temporal axis was taken to capture the most salient regions for severe AS predictions across all timepoints. We used the “inferno” colormap (<https://matplotlib.org/stable/tutorials/colors/colormaps.html>) for visualization, where pixels closer to bright yellow are highly salient for the model prediction and pixels closer to dark purple or black are negligible.

Supplemental tables

Table S1 | External performance of an automated algorithm for detection of aortic stenosis.

	New England Cohort [2016-2020]	New England Cohort [2021]	Cedars-Sinai Cohort [2018-2019]
AUROC	0.942 (0.909, 0.966)	0.978 (0.967, 0.988)	0.952 (0.941, 0.963)
AUPR	0.282 (0.197, 0.384)	0.414 (0.251, 0.589)	0.223 (0.164, 0.301)
F1 Score	0.260 (0.219, 0.303)	0.268 (0.221, 0.319)	0.259 (0.231, 0.288)
PPV	0.159 (0.133, 0.188)	0.159 (0.130, 0.192)	0.155 (0.137, 0.173)
NPV	0.995 (0.993, 0.997)	0.998 (0.997, 0.999)	0.997 (0.995, 0.998)
Sensitivity	0.702 (0.596, 0.809)	0.850 (0.700, 0.950)	0.800 (0.723, 0.877)
Specificity	0.942 (0.936, 0.949)	0.955 (0.948, 0.963)	0.932 (0.925, 0.938)
Values in parentheses represent 95% confidence intervals determined by bootstrapping. AUROC = area under the receiver operating characteristic curve; AUPR = area under the precision-recall curve; PPV = positive predictive value; NPV = negative predictive value.			

Table S2 | External performance of an automated algorithm for detection of aortic stenosis on individual PLAX videos.

	New England Cohort [2016-2020]	New England Cohort [2021]	Cedars-Sinai Cohort [2018-2019]
AUROC	0.909 (0.889, 0.928)	0.961 (0.949, 0.972)	0.924 (0.911, 0.936)
AUPR	0.206 (0.163, 0.256)	0.353 (0.252, 0.462)	0.181 (0.149, 0.226)
F1 Score	0.205 (0.184, 0.225)	0.159 (0.137, 0.181)	0.221 (0.201, 0.241)
PPV	0.121 (0.109, 0.134)	0.089 (0.077, 0.102)	0.133 (0.121, 0.145)
NPV	0.994 (0.993, 0.995)	0.998 (0.997, 0.999)	0.994 (0.993, 0.995)
Sensitivity	0.651 (0.586, 0.717)	0.745 (0.655, 0.836)	0.663 (0.605, 0.721)
Specificity	0.928 (0.924, 0.933)	0.935 (0.930, 0.940)	0.928 (0.923, 0.932)
Values in parentheses represent 95% confidence intervals determined by bootstrapping. AUROC = area under the receiver operating characteristic curve; AUPR = area under the precision-recall curve; PPV = positive predictive value; NPV = negative predictive value.			

Table S3 | External performance of an automated algorithm for detection of aortic stenosis on one randomly selected PLAX video per study.

	New England Cohort [2016-2020]	New England Cohort [2021]	Cedars-Sinai Cohort [2018-2019]
AUROC	0.926 (0.893, 0.953)	0.970 (0.954, 0.985)	0.948 (0.934, 0.961)
AUPR	0.315 (0.215, 0.427)	0.494 (0.321, 0.667)	0.219 (0.163, 0.293)
F1 Score	0.213 (0.178, 0.249)	0.204 (0.169, 0.240)	0.250 (0.223, 0.279)
PPV	0.126 (0.105, 0.148)	0.116 (0.095, 0.137)	0.148 (0.132, 0.166)
NPV	0.995 (0.993, 0.996)	0.998 (0.997, 0.999)	0.997 (0.995, 0.998)
Sensitivity	0.681 (0.574, 0.787)	0.850 (0.700, 0.950)	0.800 (0.723, 0.877)
Specificity	0.927 (0.919, 0.934)	0.936 (0.927, 0.944)	0.928 (0.921, 0.935)
Values in parentheses represent 95% confidence intervals determined by bootstrapping. AUROC = area under the receiver operating characteristic curve; AUPR = area under the precision-recall curve; PPV = positive predictive value; NPV = negative predictive value.			

Table S4 | External performance of an automated algorithm for detection of aortic stenosis when defining severe AS with objective echocardiographic cut-offs.

	New England Cohort [2021]
AUROC	0.927 (0.898, 0.952)
AUPR	0.428 (0.335, 0.530)
F1 Score	0.465 (0.396, 0.533)
PPV	0.374 (0.316, 0.436)
NPV	0.987 (0.984, 0.990)
Sensitivity	0.615 (0.523, 0.708)
Specificity	0.966 (0.959, 0.973)
In this analysis, severe aortic stenosis was defined as any study with a mean aortic valve gradient >40 mmHg, peak aortic velocity >4 m/s, or estimated aortic valve area <1 cm². Values in parentheses represent 95% confidence intervals determined by bootstrapping. AUROC = area under the receiver operating characteristic curve; AUPR = area under the precision-recall curve; PPV = positive predictive value; NPV = negative predictive value.	

Table S5 | External performance on the Cedars-Sinai cohort at different severe AS prevalence rates.

	1.5% Prevalence	5% Prevalence	10% Prevalence	15% Prevalence	20% Prevalence	Full Prevalence
Severe AS, n (%)	65 (1.5%)	221 (5.0%)	464 (10.0%)	737 (15.0%)	1043 (20.0%)	1411 (25.3%)
AUROC	0.952 (0.941, 0.963)	0.955 (0.948, 0.962)	0.951 (0.945, 0.956)	0.952 (0.947, 0.957)	0.951 (0.946, 0.956)	0.951 (0.946, 0.955)
AUPR	0.223 (0.164, 0.301)	0.483 (0.434, 0.538)	0.616 (0.579, 0.654)	0.730 (0.702, 0.758)	0.786 (0.764, 0.808)	0.827 (0.809, 0.845)
F1 Score	0.259 (0.231, 0.288)	0.515 (0.484, 0.545)	0.644 (0.619, 0.668)	0.729 (0.710, 0.748)	0.764 (0.748, 0.779)	0.791 (0.777, 0.804)
PPV	0.155 (0.137, 0.173)	0.381 (0.355, 0.408)	0.556 (0.530, 0.582)	0.674 (0.652, 0.696)	0.743 (0.724, 0.762)	0.796 (0.780, 0.812)
NPV	0.997 (0.995, 0.998)	0.988 (0.986, 0.991)	0.973 (0.969, 0.976)	0.963 (0.958, 0.967)	0.946 (0.941, 0.951)	0.928 (0.922, 0.933)
Sensitivity	0.800 (0.723, 0.877)	0.792 (0.747, 0.837)	0.765 (0.733, 0.797)	0.795 (0.771, 0.820)	0.786 (0.765, 0.806)	0.786 (0.768, 0.804)
Specificity	0.932 (0.925, 0.938)	0.932 (0.925, 0.938)	0.932 (0.925, 0.938)	0.932 (0.925, 0.938)	0.932 (0.925, 0.938)	0.932 (0.925, 0.938)
Values in parentheses represent 95% confidence intervals determined by bootstrapping. Performance is reported at various prevalence rates by maintaining all studies without severe AS in the cohort, then randomly sampling a subset of severe AS studies to achieve the desired prevalence. AUROC = area under the receiver operating characteristic curve; AUPR = area under the precision-recall curve; PPV = positive predictive value; NPV = negative predictive value.						

Table S6 | Cross-sectional association of deep learning model-derived output for severe aortic stenosis and echocardiographic parameters of diastolic dysfunction.

Independent predictor	Coefficient (95% confidence interval), p value*
Peak aortic valve velocity (cm/sec)	0.17 (0.15-0.18), p<0.001
Average E/e'	0.60 (0.37-0.83), p<0.001
Left atrial volume index (mL/m ²)	0.09 (0.02-0.17), p=0.016
Maximal tricuspid regurgitation velocity (m/sec)	0.15 (0.05-0.25), p=0.003
<i>*Results of a multivariable linear regression model with the ensemble model output prediction for severe aortic stenosis (expressed as a percentage, %) as the dependent variable. E/e': ratio of early diastolic mitral inflow velocity to early diastolic mitral annulus velocity. Findings reported in 1031 studies of the temporally distinct New England testing cohort with complete diastolic function parameters.</i>	

Supplemental figures

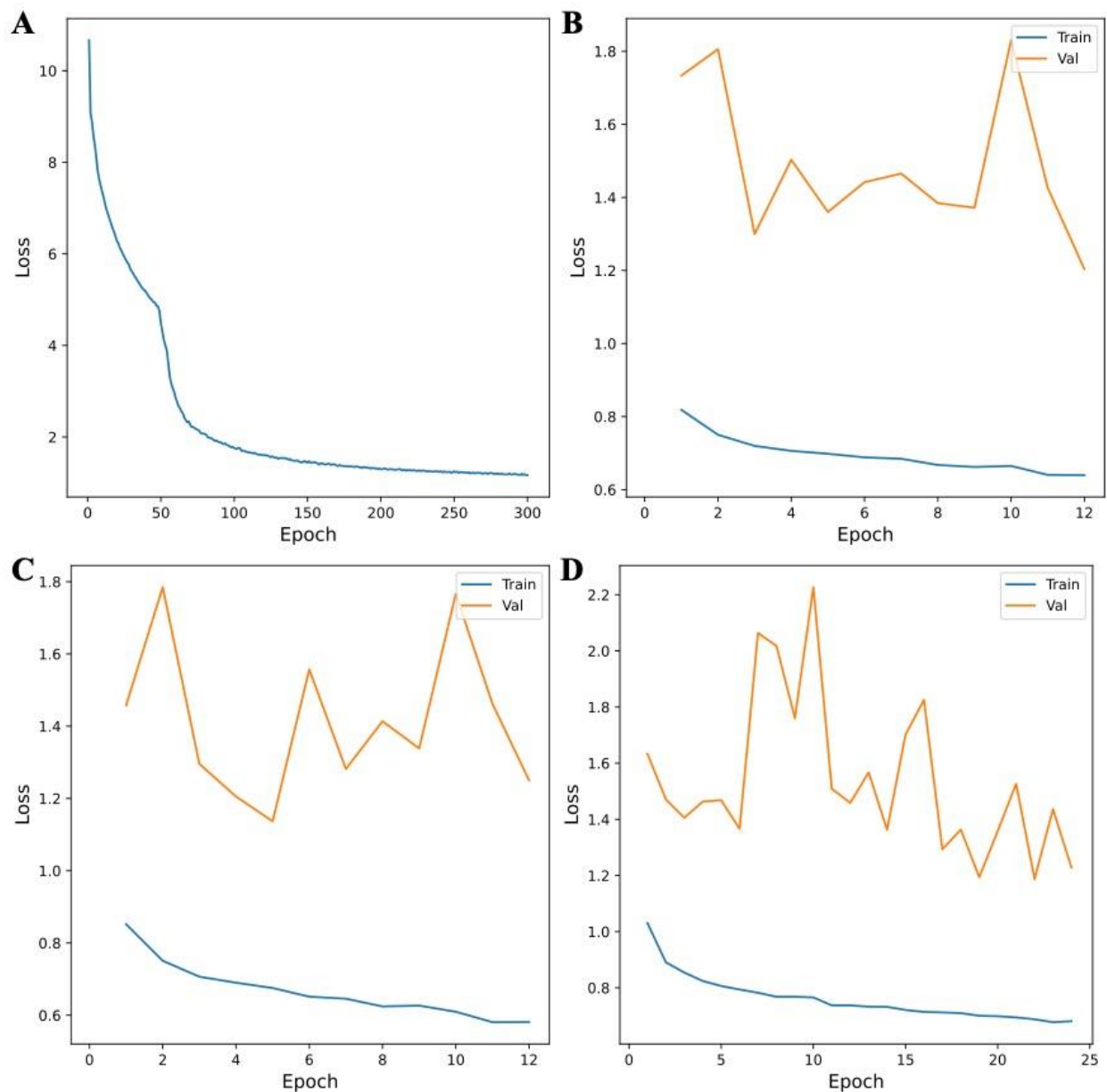


Figure S1 | Learning curves during model training. Training curves depicting model loss throughout self-supervised learning (SSL) (a), fine-tuning from a random initialization (b), fine-tuning from a Kinetics-400 initialization (c), and fine-tuning from our SSL initialization (d).

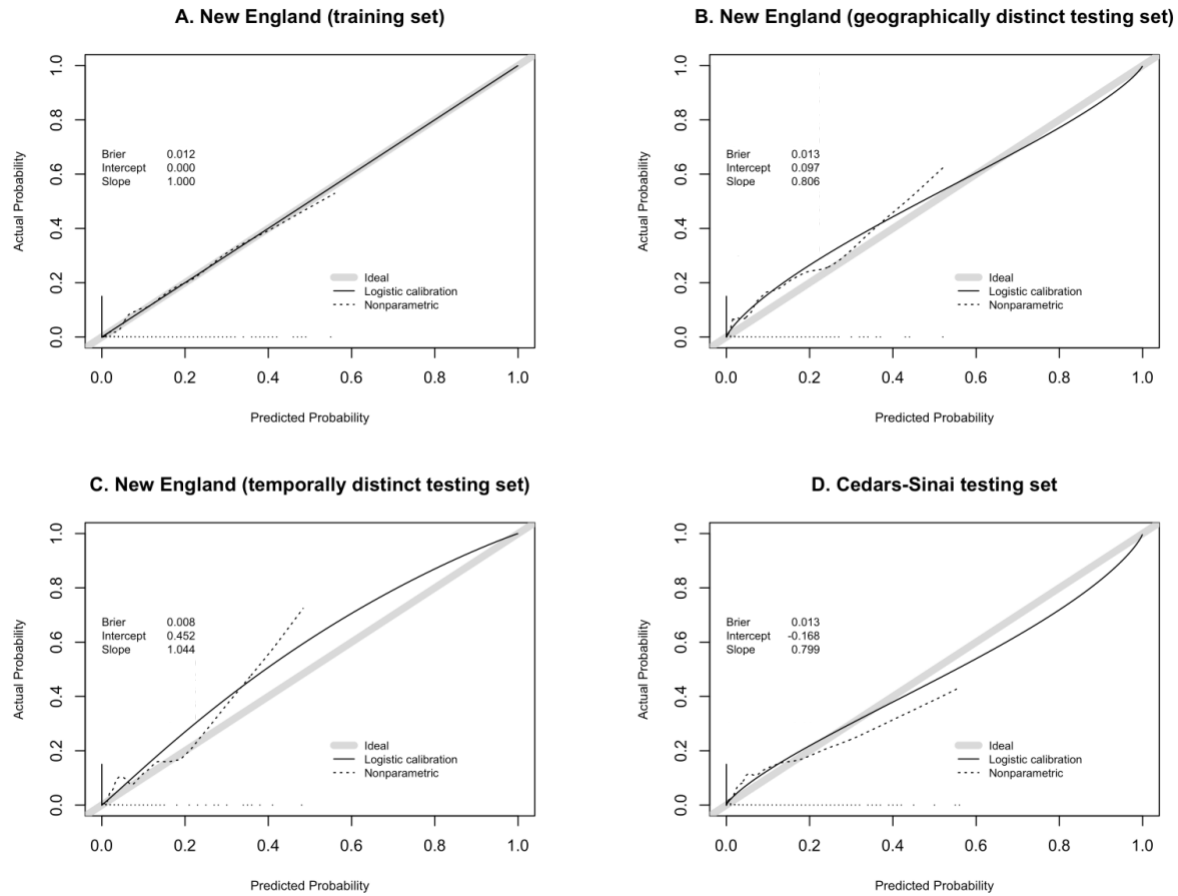


Figure S2 | Calibration performance across the training and testing sets. Calibration curves and statistics (including the Brier score, calibration slope and intercept) in (A) a down-sampled version of the New England training set (n=61 severe aortic stenosis cases among n=4093 studies; prevalence ~1.5%), (B) the geographically distinct New England testing set, (C) the temporally distinct New England testing set, and (D) the Cedars-Sinai testing set.

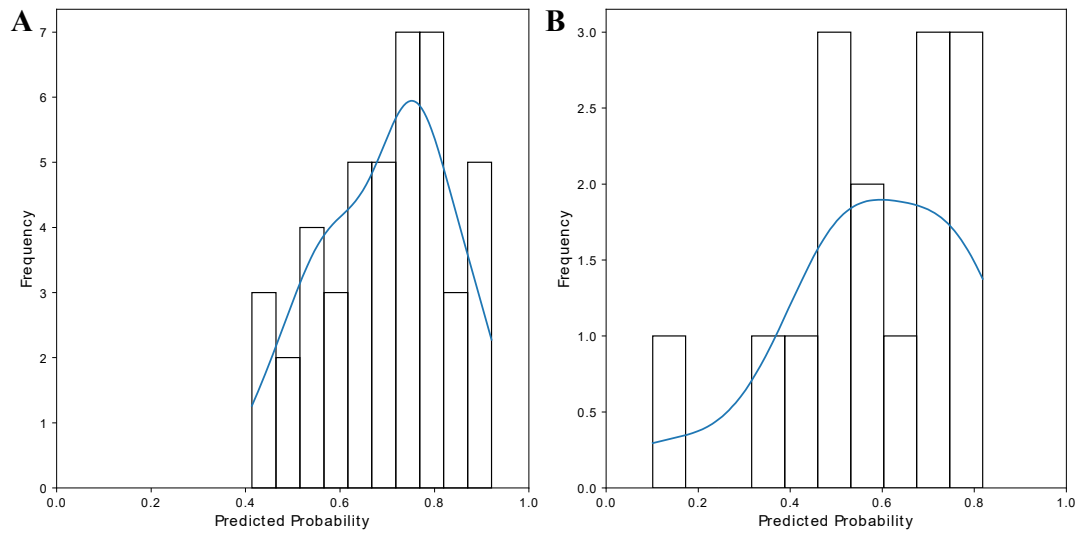


Figure S3 | Predicted model probabilities for LFLG AS studies. Histogram of predicted severe AS probabilities for the N=44 LFLG studies excluded from the New England cohorts (left) and N=15 LFLG studies excluded from the Cedars-Sinai cohort (right). LFLG = (paradoxical) low-flow, low-gradient aortic stenosis.

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